

3-Hour Daily Action Plan for JMLR Paper Submission

Period: Wednesday, January 21 - Submission Ready



CRITICAL PATH SUMMARY

Based on the assessment, you need to:

1. **Fix experimental validation** (Priority 1) - Your results don't match claims
2. **Define metrics precisely** (Priority 2) - Extrapolation error undefined
3. **Add failure case examples** (Priority 3) - Show concrete examples

Estimated timeline: 6-8 weeks → Target submission: **Mid-March 2026**



WEEK 1: Debug & Infrastructure (Jan 21-25)

Wednesday, January 21 - System Diagnosis

Goal: Identify all bugs preventing experiments from running

- **Hour 1:** Run diagnostic tests

bash

```
python standalone_real_methods_test.py --protocol comparative --verbose
```

- Document all error messages
- List which methods fail to initialize
- Check configuration files
- **Hour 2:** Fix Hybrid System v40 bug
 - Locate method name issue: `discover_validate_interpret` → `discover`
 - Update method calls in test harness
 - Verify initialization succeeds
- **Hour 3:** Fix Integrated LLM v12 bug
 - Add missing `config` parameter
 - Test initialization independently
 - Document required configuration structure

Deliverable: Bug fix report with verified initialization

Thursday, January 22 - Test Infrastructure

Goal: Get all methods running successfully

- **Hour 1:** Verify all baseline methods work
 - Pure LLM (Enhanced)
 - Neural Network baseline
 - Pure PySR
 - Test on 2-3 simple cases (Ohm's Law, Kinetic Energy)
- **Hour 2:** Verify hybrid methods work
 - Hybrid System v40
 - Integrated LLM Discovery v12
 - LLM-Guided variant
 - Test on same simple cases
- **Hour 3:** Create automated test runner

python

```
# test_runner.py

methods = [

    "Pure_LLM_Enhanced",

    "Hybrid_System_v40",

    "Integrated_LLM_v12",

    "Pure_PySR"

]

# Run all methods on test suite

# Log results to structured JSON
```

Deliverable: Working test harness for all methods

Friday, January 23 - Extrapolation Test Suite

Goal: Implement extrapolation error calculation

- **Hour 1:** Implement extrapolation data generation

python

```
def generate_extrapolation_data(X_train, formula, factor=2.0):

    """Generate test data 2x outside training range"""

    x_max = X_train.max()

    X_extrap = np.linspace(2*x_max, 4*x_max, 100)
```

```
y_extrap = formula(X_extrap)
return X_extrap, y_extrap
```

- **Hour 2:** Implement extrapolation error metric (Eq. 4 from paper)

python

```
def extrapolation_error(model, D_train, D_extrap):
    rmse_train = calculate_rmse(model, D_train)
    rmse_extrap = calculate_rmse(model, D_extrap)
    return (rmse_extrap / rmse_train) * 100
```

- **Hour 3:** Test metric on known formulas
 - Linear: should have low error (~10-20%)
 - LLM overfitting: should have high error (>100%)
 - Verify against paper claims

Deliverable: Extrapolation test module with verification

Saturday, January 24 - Ground Truth Test Suite

Goal: Select and prepare 5 ground truth tests

- **Hour 1:** Select tests (must have known analytical solutions)
 1. Ohm's Law: $V = IR$
 2. Kinetic Energy: $E = \frac{1}{2}mv^2$
 3. Ideal Gas Law: $PV = nRT$
 4. Arrhenius Equation: $k = A \exp(-E_a/RT)$
 5. Impermanent Loss: $IL = (2\sqrt{r}/(1+r) - 1) \times 100$
- **Hour 2:** Generate synthetic data for each test
 1. Add realistic noise (5-10%)
 2. Create training range $[x_{\min}, x_{\max}]$
 3. Create 3 extrapolation ranges (1.2 $\times$, 2 $\times$, 5 $\times$)
- **Hour 3:** Document ground truth formulas
 1. LaTeX format for paper
 2. Python reference implementations
 3. Expected R^2 and extrapolation errors

Deliverable: 5 ground truth test cases ready

WEEK 2: Run Core Experiments (Jan 26-Feb 1)

Monday, January 26 - Pure LLM Baseline

Goal: Get real Pure LLM results

- **Hour 1:** Run Pure LLM on 5 ground truth tests
 - Use Claude Sonnet 4 API
 - Save all outputs (formulas, code, R^2)
 - Log any failures
- **Hour 2:** Evaluate extrapolation performance
 - Calculate extrapolation error at $1.2\times$, $2\times$, $5\times$
 - Compare to ground truth
 - Document failure modes
- **Hour 3:** Analyze and record results
 - Which tests succeeded?
 - What are actual R^2 values?
 - What are actual extrapolation errors?
 - Update notes for Table 1

Deliverable: Real Pure LLM results (not hypothetical)

Tuesday, January 27 - Pure Symbolic Baseline

Goal: Get real PySR results

- **Hour 1:** Run Pure PySR on 5 ground truth tests
 - Standard configuration (from Appendix)
 - No LLM initialization
 - 500 generations per test
- **Hour 2:** Evaluate and validate
 - Calculate R^2 on training data
 - Calculate extrapolation errors
 - Check functional correctness vs. ground truth
- **Hour 3:** Record results
 - Success rate (X/5)
 - Average R^2
 - Average extrapolation error
 - Time per test
 - Update notes for Table 1

Deliverable: Real Pure PySR baseline results

Wednesday, January 28 - Hybrid System v40

Goal: Get real hybrid results

- **Hour 1:** Run Hybrid System v40 on 5 tests
 - Ensure bug fix from Jan 21 works
 - Monitor which discovery mode activates
 - Log decision points
- **Hour 2:** Evaluate performance
 - R^2 and extrapolation errors
 - Functional correctness
 - Time breakdown (LLM vs. symbolic)
- **Hour 3:** Compare to baselines
 - Better than Pure LLM?
 - Better than Pure PySR?
 - Statistical significance?
 - Update notes for Table 1

Deliverable: Real Hybrid System results

Thursday, January 29 - Integrated LLM v12

Goal: Get real integrated results

- **Hour 1:** Run Integrated LLM v12 on 5 tests
 - Ensure config fix works
 - Monitor LLM-guided initialization
 - Log warm-start success rate
- **Hour 2:** Evaluate performance
 - R^2 and extrapolation errors
 - Discovery mode breakdown
 - Compare to Hybrid System v40
- **Hour 3:** Analyze efficiency gains
 - Time savings from LLM initialization
 - Success rate improvement
 - Cost analysis (API calls)
 - Update notes for Table 1

Deliverable: Real Integrated LLM results

Friday, January 30 - Statistical Analysis

Goal: Calculate significance and confidence intervals

- **Hour 1:** Run each method 5 times on each test

- Get variance estimates
- Calculate standard deviations
- Check for outliers
- **Hour 2:** Statistical significance tests
 - Pure LLM vs. Hybrid: t-test
 - Pure PySR vs. Hybrid: t-test
 - Calculate p-values
 - 95% confidence intervals
- **Hour 3:** Prepare results summary
 - Mean $\pm$ SD for all metrics
 - Significance markers (* p<0.05, ** p<0.01)
 - Effect sizes
 - Draft updated Table 1

Deliverable: Statistical analysis complete

Saturday, January 31 - Update Paper Tables

Goal: Replace hypothetical results with real data

- **Hour 1:** Update Table 1 (Main Results)

latex

```
\begin{table}
```

```
Method & R2 & Extrap Error & Correct & Time \\\
```

```
Pure LLM & [REAL] & [REAL] & [REAL]/5 & [REAL]h \\\
```

```
Hybrid (Ours) & [REAL] & [REAL] & [REAL]/5 & [REAL]h \\\
```

- **Hour 2:** Update extrapolation table
 - Add measured 1.2 $\times$, 2 $\times$, 5 $\times$ errors
 - Verify catastrophic LLM failure claim
 - Update figure references
- **Hour 3:** Cross-check all numbers
 - Do claims match results?
 - Are confidence intervals correct?
 - Are p-values accurate?
 - Proofread table formatting

Deliverable: Updated Tables 1 & 2 with real data

WEEK 3: Failure Analysis (Feb 2-8)

Monday, February 2 - Failure Case Selection

Goal: Document 3 concrete failure examples

- **Hour 1:** Identify best failure examples
 - Silent semantic error (Impermanent Loss?)
 - Distributional reasoning (VaR?)
 - Library API misuse (Expected Shortfall?)
- **Hour 2:** Extract failure details
 - LLM output (exact code)
 - What went wrong?
 - Why did R^2 look good?
 - What was the actual impact?
- **Hour 3:** Verify with hybrid system
 - Did validation catch it?
 - What was validation score?
 - How did symbolic method fix it?

Deliverable: 3 documented failure cases

Tuesday, February 3 - Write Failure Case Study 1

Goal: Complete Impermanent Loss example

- **Hour 1:** Write case study structure

latex

`\subsubsection{Concrete Failure Example: Impermanent Loss}`

`\paragraph{Ground Truth:}`

`\paragraph{LLM Output:}`

`\paragraph{Error Analysis:}`

`\paragraph{HypatiaX Detection:}`

- **Hour 2:** Add code listings and analysis
 - Show exact LLM-generated code
 - Highlight the bug (missing $\times 100$)
 - Show $R^2 = 0.98$ (deceptive!)
 - Explain financial impact
- **Hour 3:** Add HypatiaX resolution
 - How did dimensional analysis catch it?
 - What was validation score?

- Show corrected formula
- Verify it works

Deliverable: Complete failure case study 1

Wednesday, February 4 - Write Failure Case Studies 2 & 3

Goal: Complete VaR and Expected Shortfall examples

- **Hour 1:** VaR failure case
 - Wrong sign convention
 - Show $R^2 = -10.05$
 - Explain statistical confusion
 - HypatiaX detection and fix
- **Hour 2:** Expected Shortfall failure case
 - Scipy API misuse
 - Runtime error vs. silent failure
 - Show frozen distribution problem
 - HypatiaX validation
- **Hour 3:** Polish all three case studies
 - Consistent formatting
 - Clear code highlighting
 - Verification that examples are real
 - Cross-references to experiments

Deliverable: 3 complete failure case studies

Thursday, February 5 - Add Failure Section to Paper

Goal: Integrate failure examples into Section 3

- **Hour 1:** Update Section 3.3 structure
 - Expand "Case Studies" subsection
 - Add three subsections (3.3.1, 3.3.2, 3.3.3)
 - Ensure flow from motivation
- **Hour 2:** Add figures/tables
 - Table: Failure mode taxonomy
 - Code listings with highlighting
 - Validation score comparisons
- **Hour 3:** Connect to theory
 - Reference Theorem 2.2
 - Explain why LLMs make these errors
 - Link to extrapolation failure

- Preview hybrid solution

Deliverable: Expanded Section 3 with failure analysis

Friday, February 6 - Define Extrapolation Metric

Goal: Add formal metric definition to paper

- **Hour 1:** Write Section 5.2 "Extrapolation Error Metric"

latex

```
\subsection{Extrapolation Error Metric}
```

We define extrapolation capability as...

[Add Equation 4]

- **Hour 2:** Add figure showing extrapolation
 - Plot error vs. distance from training
 - Show LLM catastrophic failure
 - Show hybrid graceful degradation
 - Add error bars
- **Hour 3:** Connect to main results
 - Reference in Section 5.3
 - Explain why metric matters
 - Link to Table 1 results
 - Proofread equations

Deliverable: Formal extrapolation metric definition

Saturday, February 7 - Generate Figures

Goal: Create publication-quality figures

- **Hour 1:** Figure 1 - Architecture diagram
 - Already exists, verify quality
 - Update if needed for clarity
 - Ensure all layers labeled
- **Hour 2:** Figure 2 - Extrapolation comparison

python

```
plt.figure(figsize=(10, 6))  
plt.plot(distances, llm_errors, 'r-', label='Pure LLM')  
plt.plot(distances, hybrid_errors, 'b-', label='Hybrid')
```

- Show 1.2×, 2×, 5× ranges
- Add error bars
- Clear legend and labels
- **Hour 3:** Figure 3 - Discovery mode breakdown
 - Pie chart or bar chart
 - LLM-only: 73%
 - LLM+PySR: 23%
 - PySR fallback: 4%

Deliverable: 3 publication-quality figures



WEEK 4: Polish & Internal Review (Feb 9-15)

Monday, February 9 - Abstract & Introduction

Goal: Ensure consistency with results

- **Hour 1:** Update abstract
 - Replace "412-847%" with REAL numbers
 - Update success rates with REAL data
 - Verify all claims match experiments
- **Hour 2:** Update introduction
 - Section 1.1: Update motivating evidence
 - Verify bimodal behavior claim
 - Check extrapolation numbers
- **Hour 3:** Update contributions list
 - Ensure all 4 contributions delivered
 - Check cross-references
 - Verify no overclaiming

Deliverable: Updated abstract & introduction

Tuesday, February 10 - Results Section Review

Goal: Ensure Section 5 is complete and accurate

- **Hour 1:** Review Section 5.3 (Main Results)
 - Verify Table 1 numbers
 - Check all cross-references
 - Ensure statistical tests included
- **Hour 2:** Review Section 5.4 (Domain Performance)
 - Verify Table 2 numbers
 - Check domain-specific analysis

- Update any outdated text
- **Hour 3:** Review Section 5.5 (Failure Analysis)
 - Verify consistency with case studies
 - Check that validation caught failures
 - Ensure conclusions justified

Deliverable: Polished Section 5

Wednesday, February 11 - Discussion & Conclusion

Goal: Update Sections 6 & 7 based on real results

- **Hour 1:** Update Section 6.1 (Principal Findings)
 - Replace hypothetical with real numbers
 - Verify theoretical claims supported
 - Check Theorem 2.2 connection
- **Hour 2:** Update Section 6.2 (Design Principles)
 - Ensure principles reflect experiments
 - Add any new insights
 - Verify actionability
- **Hour 3:** Update Section 7 (Conclusion)
 - Replace all numbers with real results
 - Verify no overclaiming
 - Check that limitations honest
 - Ensure future work realistic

Deliverable: Updated Sections 6 & 7

Thursday, February 12 - Appendix & Reproducibility

Goal: Ensure reproducibility materials complete

- **Hour 1:** Update Appendix C (Detailed Results)
 - Replace with real experimental data
 - Add session timestamps
 - Verify checksums if needed
- **Hour 2:** Update Appendix D (Reproducibility)
 - Verify GitHub links work
 - Check Docker container
 - Test quick start commands
 - Update expected runtimes with real data
- **Hour 3:** Update Appendix E (Complexity Analysis)
 - Verify timing breakdowns match experiments

- Update scalability table
- Check computational cost claims

Deliverable: Complete appendices

Friday, February 13 - Internal Consistency Check

Goal: Ensure entire paper is self-consistent

- **Hour 1:** Cross-reference audit
 - Check all `\ref{}` commands work
 - Verify all `\cite{}` commands valid
 - Check figure/table numbers
- **Hour 2:** Number verification
 - List all quantitative claims
 - Verify each against experimental data
 - Flag any inconsistencies
- **Hour 3:** Claim verification
 - List all major claims
 - Verify evidence for each
 - Ensure no unsupported claims

Deliverable: Consistency audit report

Saturday, February 14 - Fresh Eyes Proofread

Goal: Catch errors and improve clarity

- **Hour 1:** Read abstract + introduction
 - Mark unclear sentences
 - Check flow and logic
 - Note any jargon
- **Hour 2:** Read methods + results
 - Verify technical accuracy
 - Check equation formatting
 - Note any gaps
- **Hour 3:** Read discussion + conclusion
 - Check argument flow
 - Verify conclusions supported
 - Note any overclaiming

Deliverable: Proofread notes



WEEK 5: External Review (Feb 16-22)

Monday, February 16 - Prepare Review Package

Goal: Package paper for external reviewers

- **Hour 1:** Create review checklist
 - Technical accuracy
 - Clarity and readability
 - Completeness of experiments
 - Statistical rigor
 - Contribution assessment
- **Hour 2:** Identify 2-3 reviewers
 - 1 symbolic regression expert
 - 1 LLM researcher
 - 1 scientific ML practitioner
- **Hour 3:** Send review requests
 - Include paper PDF
 - Include checklist
 - Request feedback by Feb 22
 - Offer co-authorship if substantial

Deliverable: Review requests sent

Tuesday-Thursday (Feb 17-19) - Wait for Feedback

Use this time for:

- Backup experiments (run more test cases)
 - Supplementary material preparation
 - Bibliography verification
 - LaTeX formatting polish
-

Friday, February 20 - Address Reviewer 1 Feedback

Goal: Incorporate first reviewer's comments

- **Hour 1:** Catalog feedback
 - List all substantive comments
 - Prioritize by importance
 - Identify quick fixes
- **Hour 2:** Address technical comments
 - Fix any errors
 - Clarify ambiguities

- Add missing details
- **Hour 3:** Update affected sections
 - Track changes in document
 - Verify fixes complete
 - Thank reviewer

Deliverable: Reviewer 1 feedback addressed

Saturday, February 21 - Address Reviewers 2 & 3 Feedback

Goal: Incorporate remaining feedback

- **Hour 1:** Address Reviewer 2
 - Similar process as Friday
 - Track all changes
- **Hour 2:** Address Reviewer 3
 - Similar process
 - Resolve any conflicts between reviewers
- **Hour 3:** Final integration
 - Ensure all changes compatible
 - Re-proofread modified sections
 - Update version number

Deliverable: All reviewer feedback addressed



WEEK 6: Final Polish (Feb 23-Mar 1)

Monday, February 23 - LaTeX Formatting

Goal: Perfect formatting for JMLR

- **Hour 1:** Verify JMLR style compliance
 - Check jmlr2e.sty usage
 - Verify citation format
 - Check figure/table placement
- **Hour 2:** Polish equations
 - Consistent notation
 - Proper alignment
 - Clear equation numbers
- **Hour 3:** Polish code listings
 - Syntax highlighting
 - Proper indentation
 - Caption clarity

Deliverable: Formatted to JMLR standards

Tuesday, February 24 - Bibliography Audit

Goal: Ensure all citations complete and accurate

- **Hour 1:** Check all BibTeX entries
 - Complete author lists
 - Correct venues
 - Accurate years
 - DOIs where available
- **Hour 2:** Verify citation context
 - Each cite supports claim
 - No misleading citations
 - Proper attributions
- **Hour 3:** Check for missing citations
 - Standard references included
 - Recent work (2024-2025) cited
 - No self-plagiarism

Deliverable: Complete bibliography

Wednesday, February 25 - Generate Supplementary Materials

Goal: Prepare supplementary document

- **Hour 1:** Extract additional results
 - Extended test cases
 - Additional failure examples
 - Hyperparameter sensitivity
- **Hour 2:** Format supplementary PDF
 - Clear section structure
 - References to main paper
 - Self-contained explanations
- **Hour 3:** Verify code repository
 - All scripts work
 - README complete
 - Data available
 - License specified

Deliverable: Supplementary materials ready

Thursday, February 26 - Final Internal Review

Goal: Last check before submission

- **Hour 1:** Complete paper read-through
 - Check overall narrative
 - Verify logical flow
 - Note any final issues
- **Hour 2:** Checklist verification
 - Abstract accurate?
 - All tables real data?
 - All figures clear?
 - Equations correct?
 - Citations complete?
 - Reproducibility materials ready?
 - Supplementary materials ready?
- **Hour 3:** Final fixes
 - Address any issues found
 - Last typo check
 - Compile final PDF

Deliverable: Submission-ready draft

Friday, February 27 - Prepare JMLR Submission

Goal: Package everything for submission

- **Hour 1:** Prepare submission files
 - Main PDF
 - LaTeX source files
 - Figures (separate files)
 - Supplementary PDF
 - Code repository link
- **Hour 2:** Write cover letter
 - Summarize contributions
 - Explain significance
 - Suggest reviewers
 - Declare conflicts of interest
- **Hour 3:** Complete JMLR forms
 - Author information
 - Keywords
 - Copyright agreement
 - Reproducibility checklist

Deliverable: Complete submission package

Saturday, February 28 - Submit to JMLR

Goal: Submit paper

- **Hour 1:** Upload to JMLR portal
 - Create account if needed
 - Upload all files
 - Complete metadata
- **Hour 2:** Verify submission
 - Check uploaded PDF renders correctly
 - Verify all files included
 - Review submission details
- **Hour 3:** Post-submission tasks
 - Archive submission version
 - Post preprint to arXiv
 - Update personal website
 - Celebrate! 🎉

Deliverable: Paper submitted!



SUMMARY TIMELINE

Week	Focus	Deliverable
Week 1 (Jan 21-25)	Debug & Infrastructure	Working test harness
Week 2 (Jan 26-Feb 1)	Run Core Experiments	Real experimental results
Week 3 (Feb 2-8)	Failure Analysis	Documented failure cases
Week 4 (Feb 9-15)	Polish & Internal Review	Self-consistent draft
Week 5 (Feb 16-22)	External Review	Reviewer feedback incorporated
Week 6 (Feb 23-Mar 1)	Final Polish & Submit	Submitted to JMLR



CRITICAL CHECKPOINTS

End of Week 2 (Feb 1):

- ☒ All 5 methods run successfully
- ☒ Real data in Tables 1 & 2
- ☒ If fails: Extend Week 2 by 3-5 days

End of Week 4 (Feb 15):

- ☒ Paper internally consistent
- ☒ All claims supported by data

- ❌ If fails: Do NOT proceed to external review

End of Week 5 (Feb 22):

- ✅ External feedback incorporated
 - ✅ No major issues remaining
 - ❌ If fails: Add 1 week for revisions
-

SUCCESS CRITERIA

By submission date (Feb 28), you must have:

1. ✅ **Real experimental results** matching paper claims
 2. ✅ **Defined extrapolation error** with verification
 3. ✅ **Documented failure cases** with concrete examples
 4. ✅ **Statistical significance tests** (p-values, CIs)
 5. ✅ **Complete reproducibility materials** (code, data, docs)
 6. ✅ **External review** from 2-3 domain experts
 7. ✅ **Self-consistent narrative** with no contradictions
-



EFFICIENCY TIPS

1. **Work in order** - Don't skip ahead, each week builds on previous
2. **Track progress** - Check off items as completed
3. **Stop if blocked** - If experiments fail, fix immediately
4. **Batch similar tasks** - All coding in morning, all writing in afternoon
5. **Use weekends** - For longer runs (experiments, reviews)

Total effort: ~126 hours over 6 weeks (18 days × 3 hours + weekends)

Good luck! 🚀